Edeviser — Complete Product Documentation

For Investment Preparation & Scalability Planning

Version: 2.1 | **Date:** April 2026 | **Purpose:** Comprehensive product reference for investor Q&A, due diligence, and strategic planning

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1. EXECUTIVE SUMMARY

What is Edeviser?

Edeviser is a **Human-Centric Outcome-Based Education (OBE) + Gamification platform** for higher education institutions. It resolves the paradox between accreditation compliance and student engagement through a **Dual-Engine Architecture**:

- **Engine 1 — OBE Core:** Automates ILO → PLO → CLO mapping, rubric-based grading, immutable evidence generation, and accreditation report export.
- **Engine 2 — Habit Core:** Transforms compliance activities into motivating student experiences via XP, streaks, badges, levels, leaderboards, AI tutoring, and self-regulated learning tools.

Key Differentiator

Where competitors treat compliance and engagement as separate products, Edeviser fuses them into a single feedback loop — *compliance generates gamified triggers; student engagement generates accreditation evidence.*

Vision

> "*To make every learning interaction count — for the student who needs to grow, and the institution that needs to prove it.*"

Category

EdTech SaaS (B2B2C) — Higher Education Accreditation & Student Engagement Platform

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2. USER PERSONAS & ROLES

2.1 Administrator

- System owner, compliance officer, IT manager
- Core need: Generate comprehensive, auditable outcome attainment reports instantly
- Pain: Data siloed across spreadsheets; last-minute accreditation preparation
- Success: One-click PDF export of institution-wide PLO attainment with evidence citations

2.2 Coordinator (Program Manager)

- Curriculum architect, program oversight
- Core need: Live visual matrix of which courses cover which PLOs
- Pain: No real-time visibility; mapping errors discovered only at audit
- Success: Live, color-coded PLO × Course matrix with drill-down to evidence

2.3 Teacher (Faculty)

- Instructor, content creator, assessor
- Core need: Grade once → evidence auto-generated; no separate OBE form

- Pain: Dual data entry; rubric creation complexity; no real-time student insight
- Success: 5-minute assignment setup with rubric templates; automatic evidence chain

2.4 Student

- Primary end-user, digital native
- Core need: Feel immediate progress and understand skill connections
- Pain: No feedback loop; grades arrive late; feels like a number
- Success: Real-time XP animation on submission; visible skill progression; 30+ day streaks

2.5 Parent/Guardian

- Read-only dashboard for linked student grades, attendance, CLO progress, and habit data

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3. THE 10 FOUNDATIONAL PILLARS (Science-Backed)

Engine 1: OBE Core (Structure & Compliance)

Pillar	Science Basis	Product Manifestation
1. Outcome-Based Education	Spady (1994)	ILO → PLO → CLO mapping; automated evidence rollup
2. Rubrics	Andrade (2000)	Per-CLO rubric builder with level descriptors
3. Bloom's Taxonomy	Bloom (1956), Anderson & Krathwohl (2001)	CLO tagging; visual distribution charts; Learning Path ordering

Engine 2: Habit Core (Behavior & Motivation)

Pillar	Science Basis	Product Manifestation
4. Agentic AI	LLMs + RAG	AI Tutor, at-risk detection, feedback drafts, adaptive quizzes
5. Self-Regulated Learning	Zimmerman (1989)	Personal dashboard, habit tracking, goal setting, weekly planner
6. BJ Fogg Behavior Model	Fogg (2009)	Daily micro-task prompts; graduated habit difficulty levels
7. Hooked Model	Nir Eyal (2014)	Daily login triggers; variable XP rewards; leaderboard investment
8. Gamification (Octalysis)	Yu-kai Chou	XP, levels, badges, streaks, leaderboards, team challenges
9. Flow Theory	Csikszentmihalyi (1990)	Adaptive difficulty; Focus Mode; flow check-ins
10. Reflection Journaling	Kolb (1984)	CLO-linked journal prompts; Gibbs' Reflective Cycle templates

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4. TECH STACK & ARCHITECTURE

Frontend

- React 18 + TypeScript (strict mode), Vite 6, Tailwind CSS v4 + Shadcn/ui
- React Router v7, TanStack Query v5, TanStack Table, React Hook Form + Zod
- Framer Motion, Recharts, dnd-kit, Sonner, nuqs, canvas-confetti
- Zustand, i18next + react-i18next, Sentry

Backend (Supabase BaaS)

- PostgreSQL 15 with Row Level Security (RLS) on ALL tables
- GoTrue Auth (JWT, bcrypt, token refresh)
- Supabase Realtime (Phoenix Channels) for live updates
- Supabase Storage (file uploads with RLS)
- Supabase Edge Functions (Deno runtime) for complex logic
- pgvector for AI RAG pipeline, pg_cron for scheduled jobs

Deployment

- Vercel (frontend CDN, CI/CD, preview deployments)
- Supabase Cloud (managed PostgreSQL, Auth, Storage, Edge Functions)

Architecture Pattern

Backend-as-a-Service (BaaS) + Edge Functions. Security enforced at the database layer via RLS policies. Single Supabase project with RLS for multi-tenancy (revisit at 50+ institutions).

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5. COMPLETE FEATURE SET

5.1 Authentication & Authorization (LIVE)

- Email/password login via Supabase Auth with JWT
- Role-Based Access Control: admin, coordinator, teacher, student, parent
- 5-attempt lockout (15 min) with audit logging
- Role-aware post-login redirect, session persistence (24h active, 8h idle timeout)
- Password reset via email (1-hour expiry, single-use)

5.2 User & Profile Management (LIVE)

- Admin user provisioning (CRUD, soft-delete preserving evidence)
- Bulk CSV import (up to 1000 users, atomic, email invitations)
- Profile management (name, avatar 2MB limit, notification preferences)
- Admin impersonation mode (read-only, 30-min auto-expiry, audit logged)

5.3 Program & Course Management (LIVE)

- Program CRUD with coordinator assignment
- Course CRUD with teacher assignment, semester scoping
- Course sections (A, B, C) sharing CLOs with separate enrollments
- Student enrollment (active, completed, dropped states)
- Department management, semester management with date ranges

5.4 OBE Engine — The Compliance Core (LIVE)

- ILO/PLO/CLO Management with Bloom's Taxonomy tagging, drag-reorder, audit logging
- Sub-CLOs for granular skill tracking
- Outcome Mappings with weighted relationships (0-1.0), weight sum validation

- Rubric Builder: per-CLO rubrics, criteria × performance levels, reusable templates
- Assignment Creation: CLO-linked (1-3 CLOs with weights), rubric-linked, due dates, late windows
- Student Submission with file upload, late flagging, XP on submit
- Rubric-Based Grading: click-to-grade, auto-score, per-criterion feedback
- Automatic Evidence Generation: immutable append-only records (<500ms)
- Outcome Attainment Rollup: CLO → PLO → ILO weighted aggregation (<500ms)
- Curriculum Matrix: PLO × Course grid with color-coded attainment, CSV export
- Accreditation Report Export: PDF generation (jspdf), configurable templates (ABET/HEC/AACSB)
- Course File Generation, CQI Action Plans

5.5 Gamification Engine — The Engagement Core (LIVE)

- XP System: 11+ XP sources (login, submission, grading, journal, streaks, quizzes, study sessions, etc.)
- Adaptive XP Engine: dynamic adjustment based on level, difficulty, diminishing returns
- Streak System: multi-layered recovery (Comeback Challenge, Sabbatical, Streak Freeze)
- Badge System: standard, mystery, tiered (Bronze/Silver/Gold), Badge Spotlight, team badges
- Level System: 20 levels, progressive thresholds, full-screen level-up animations
- Leaderboard System: individual, Personal Best, Most Improved, League Tiers, Percentile Bands, Team Leaderboard
- Daily Habits: 4 academic habits with graduated difficulty levels (BJ Fogg), Perfect Day nudges

5.6 Habit Heatmap & Wellness (LIVE)

- Semester-long GitHub-style heatmap, 4 optional wellness habits (private by default)
- Habit Analytics Dashboard, correlation insights, level-aware rendering
- Streak recovery visualization, wellness micro-guidance, exportable reports

5.7 AI Co-Pilot (LIVE)

- Personalized Module Suggestions, At-Risk Early Warning, AI Feedback Drafts, AI Feedback Flywheel

5.8 AI Chat Tutor with RAG (PLANNED)

- Conversational AI tutoring grounded in course materials via pgvector
- Multi-persona (Socratic Guide, Step-by-Step Coach, Quick Explainer)
- Autonomy levels L1-L3, academic integrity guardrails, independence nudges
- Source citations, teacher handoff, learning plan updates
- Rate limiting: 50 messages/day, 50K tokens/day per student

5.9 Adaptive Quiz Generation (LIVE)

- AI-powered question generation via RAG, teacher review/approval workflow
- Per-student adaptive difficulty, Item Response Theory calibration
- Bloom's Climb mechanic, Mastery Recovery Pathways, Practice Mode
- AI explanation confidence indicators, post-quiz review with AI explanations

5.10 Student Onboarding & Profiling (LIVE)

- Progressive profiling: Day 1 minimal (7 questions, <3 min) → micro-assessments over 2 weeks
- Big Five, VARK (self-awareness only), Self-Efficacy Scale, Study Strategy Inventory
- Baseline diagnostic tests, AI-generated Starter Week Plan, SMART goal suggestions

5.11 Weekly Planner & Today View (LIVE)

- PDCR cycle, Focus Mode (Pomodoro + custom timer), session evidence capture
- Session intent setting, flow state check-ins, spaced repetition scheduling
- Structured reflection templates, AI reflection synthesis, journal quality scoring

5.12 Team Challenges & Social Quests (LIVE)

- Teams (2-6 students), team XP/streaks/badges, Social Challenges (4 types)
- Contribution accountability, Cooperation Score, Peer Teaching Moments
- AI-powered Team Health Monitoring

5.13 XP Marketplace & Virtual Economy (PLANNED)

- Virtual Wallet, 3 item categories (Cosmetic, Educational Perks, Power-ups)
- Dynamic pricing, XP Economist dashboard, Knowledge Quests, Mystery Reward Boxes

5.14 i18n & RTL Support (LIVE)

- Full Arabic/English bilingual, RTL layout, bilingual entity content
- Cognitive accessibility, neurodivergent support (dyslexia font, reduced animations)

5.15 Institutional Management (LIVE)

- Semesters, course sections, surveys, CQI, course file generation
- Announcements, course modules/materials, discussion forums, attendance
- Quiz/exam module, gradebook, calendar, timetable, departments
- Academic calendar, transcripts, parent portal, fee management
- Graduate attributes, competency frameworks

5.16 Advanced Visualizations (LIVE)

- Sankey diagram, coverage heatmap, gap analysis, cohort comparison, semester trends

5.17 Platform Infrastructure (LIVE)

- Global search (Cmd+K), dark mode, PWA, offline resilience, draft auto-save
- Notification system (real-time + email + batching), audit logging, activity logging
- Rate limiting, image compression, GDPR data export, session management

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6. DATABASE SCHEMA OVERVIEW

Core Tables (30+)

`institutions`, `profiles`, `programs`, `courses`, `student_courses`, `learning_outcomes`, `outcome_mappings`, `rubrics`, `rubric_criteria`, `assignments`, `submissions`, `grades`, `evidence`, `outcome_attainment`, `student_gamification`, `badges`, `xp_transactions`, `journal_entries`, `habit_logs`, `wellness_habit_logs`, `notifications`, `audit_logs`, `student_activity_log`, `ai_feedback`, `bonus_xp_events`, `semesters`, `departments`, `course_sections`, `surveys`, `quizzes`, `quiz_attempts`, `question_bank`, `question_analytics`, `teams`, `team_members`, `social_challenges`, `challenge_progress`, `study_sessions`, `planner_tasks`, `weekly_goals`, `session_evidence`, `student_profiles`, `onboarding_responses`, `baseline_attainment`, and more.

Key Design Decisions

1. Unified outcome table — ILOs, PLOs, CLOs share one table discriminated by `type`
2. Immutable evidence — Append-only, never updated or deleted
3. XP ledger pattern — Every change is a transaction record; `xp_total` is materialized sum
4. RLS on ALL tables — Security enforced at database layer
5. Single-tenant with RLS — One Supabase project, institution isolation via RLS

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7. SECURITY ARCHITECTURE

- Row Level Security on every table (no exceptions)
- JWT verification on all API endpoints
- FERPA & GDPR compliance design
- Input validation with Zod (client + server)
- Rate limiting: 5 login attempts/15 min per IP; 100 read/30 write per min on Edge Functions
- Edge Function secrets management (never client-side)
- No student PII sent to LLM providers
- Audit trail for all admin actions
- Dependency vulnerability scanning (Dependabot + Snyk)

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8. NON-FUNCTIONAL REQUIREMENTS

Performance

Metric	Target
Dashboard load (4G, cold cache)	≤ 1.5s
Evidence rollup	≤ 500ms
API response p95	≤ 300ms
Concurrent users	5,000 without degradation
AI Tutor first token	≤ 3s

Reliability

Metric	Target
Uptime	99.9%
RTO	< 4 hours
RPO	< 1 hour
Graceful degradation	AI features degrade; core OBE never blocked

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9. SUCCESS METRICS & KPIs

Product Health

Metric	Target
Student DAU/MAU Ratio	≥ 60%
Average Session Length	≥ 8 minutes
7-Day Streak Continuation	≥ 40%
On-Time Submission Rate	≥ 75%

Evidence Coverage	100% of graded assignments
Student NPS	≥ 40
Teacher NPS	≥ 30

Business KPIs

Metric	Target
Institutions Onboarded	5 in year 1
Student MAU	10,000 by end of year 1
Institution Churn Rate	< 10% annually

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10. RELEASE ROADMAP — Qatar Market Launch

Starting April 2026, targeting Qatar higher education market readiness within 6 months.

Phase	Timeline	Focus	Key Deliverables	Exit Criteria
Alpha (Internal)	Apr 2026	Backend stability & end-to-end testing	Full backend integration testing, RLS policy audit, Edge Function load testing, database migration validation, seed data for all roles, CI/CD pipeline hardening	All 25+ Edge Functions passing, zero critical RLS gaps, <500ms evidence rollup under load
Design & Market Prep	May 2026	UI/UX polish for Qatar market	Arabic RTL layout finalization, Qatar-specific accreditation templates (QU/CNA-Q), design system audit against Figma, responsive testing across devices, accessibility compliance pass	RTL parity with LTR, all pages passing responsive screenshots, Arabic translations reviewed by native speaker
Pilot v1	Jun 2026	First institutional deployment	Deploy to 1 Qatar university (1-2 departments, 200-500 students), onboard admin/coordinator/teacher users, collect structured feedback via in-app surveys, monitor performance metrics	Pilot institution actively using OBE + gamification core, feedback collected from all 4 roles
Iteration	Jul 2026	Feedback-driven refinement	Address top 10 pilot feedback items, performance optimization based on real usage data, UX friction points resolved, Arabic translation corrections, teacher training materials created	Pilot institution satisfaction score ≥ 7/10, critical feedback items resolved

Pilot v2 (Revised)	Aug 2026	Expanded pilot with AI features	Expand to 2-3 departments (1,000-2,000 students), enable AI Co-Pilot features (at-risk, suggestions, feedback drafts), activate adaptive quizzes, stress test Realtime at scale	AI features operational, system stable at 2,000 concurrent users, teacher adoption rate ≥ 60%
Beta	Sep 2026	Pre-production hardening	Security penetration testing, GDPR/data residency compliance audit, disaster recovery drill, load testing at 5,000 concurrent users, SLA documentation, onboarding 2nd pilot institution	Passing security audit, documented SLA, 2 institutions live, <1% error rate
Licensing & Contracts	Oct 2026	Commercial readiness	Finalize pricing model, prepare institutional licensing agreements, SLA contracts, data processing agreements (DPA), support tier definitions, sales collateral	Signed contracts with pilot institutions, pricing validated
Full-Scale Deployment	Nov 2026	Production launch	Full rollout to contracted institutions, 24/7 monitoring setup, support ticketing system, onboarding playbook for new institutions, marketing for Qatar/GCC expansion	3-5 institutions live, support SLA operational, pipeline for next 10 institutions

Parallel Scalability Track (Running Alongside Pilots)

Month	Scalability Work
May-Jun	Database indexing optimization, connection pooling (PgBouncer), materialized views for dashboards
Jul-Aug	Redis integration for leaderboard caching, CDN optimization, Edge Function cold-start reduction
Sep-Oct	Multi-region Supabase evaluation, read replica setup for analytics, automated backup verification
Nov onward	Architecture review for 50+ institution scale, evaluate dedicated compute tiers

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11. SCALABILITY COST ANALYSIS

Option A: Current Stack (Supabase + Vercel) — Realistic Monthly Costs

All prices based on published pricing as of April 2026. Sources: supabase.com/pricing, vercel.com/pricing, openrouter.ai, resend.com/pricing, openai.com/api/pricing, github.com/pricing.

Scale Tier	Users	Supabase	Vercel	AI (LLM)	AI (Embeddings)	Email (Resend)	Monitoring
Pilot (1 institution)	500	$25 (Pro) + $25 compute = $50	$20 (Pro)	$60-80	$2-5	$20 (Pro: 50K/mo)	$29 (Sentry)
Early Growth (3 institutions)	2,000	$25 + $50 compute + $20 storage = $95	$20 + ~$30 overages = $50	$150-250	$10-20	$20 (Pro: 50K emails)	$29
Growth (10 institutions)	10,000	$599 (Team) + $100 compute = $699	$20 + ~$80 overages = $100	$400-700	$30-50	$90 (Scale: 100K emails)	$29
Scale (30 institutions)	30,000	$599 + $300 compute + $50 storage = $949	$100 + ~$150 overages = $250	$1,000-1,800	$80-120	$90 + overages = $150	$79 (Sentry Team)
Enterprise (50+ institutions)	50,000+	Enterprise (custom, est. $2,000-3,500)	Enterprise (custom, est. $500-1,000)	$2,500-4,000	$150-250	$200-400	$79

Development Tooling Costs (Fixed Monthly)

Tool	Plan	Cost/mo	Purpose
Claude Code (Anthropic)	Max plan	$200	AI-assisted code generation, refactoring, debugging, and architecture decisions
GitHub Enterprise Cloud	2 users ($21/user)	$42	Source control, CI/CD, advanced security, includes Copilot
CodeRabbit	Pro (per seat)	$19	Automated PR code review with AI — catches bugs, security issues, and style violations before merge
Figma	Professional (2 editors)	$30 ($15/editor)	UI/UX design, component library, design-to-code handoff, responsive prototyping
Postman	Basic (1 seat)	$14	API testing, collection management, automated test suites for Edge Functions

Kiro IDE	Max	$39	Spec-driven development, steering files, agent hooks, MCP powers for CI automation
Dev Tooling Subtotal		**$344**	

Note: Sentry ($29/mo) is counted in the Monitoring column of the main cost table, not in Dev Tooling. Dev tooling costs are fixed regardless of user count. At pilot stage they represent ~63% of total spend, but at 10K+ users they drop to under 18% — a healthy ratio for a product-led engineering team. These tools directly reduce the need for additional engineering hires by accelerating development velocity 3-5x.

AI Cost Breakdown (Per 10,000 Active Students/Month)

AI Feature	Model	Est. Token Usage/mo	Cost/mo
At-Risk Predictions	GPT-4o via OpenRouter ($2.50/$10 per 1M tokens)	~20M input + 4M output tokens	$90.00
Module Suggestions	GPT-4o	~15M input + 3M output	$67.50
Feedback Drafts	GPT-4o	~40M input + 10M output	$200.00
Quiz Generation	GPT-4o	~20M input + 5M output	$100.00
AI Tutor (when live)	GPT-4o / DeepSeek	~100M input + 25M output	$500.00
Embeddings (RAG)	text-embedding-3-small/large ($0.02-$0.13/1M tokens)	~50-150M tokens	$1.00-$19.50
Total AI per 10K students			**~$958-977/mo**

Note: DeepSeek models via OpenRouter are significantly cheaper (~$0.14/$0.28 per 1M tokens) and can reduce AI Tutor costs by 80-90% with acceptable quality for educational use cases.

Supabase Cost Details

- Pro plan: $25/mo base + usage overages
- Team plan: $599/mo (SOC2, SSO, priority support — needed for enterprise clients)
- Compute add-ons: $50-300/mo depending on instance size (2 vCPU/4GB RAM = ~$50, 4 vCPU/8GB = ~$100)
- Storage: $0.021/GB beyond included 100GB (Pro) or 200GB (Team)
- Bandwidth: $0.09/GB beyond included 250GB
- Edge Function invocations: 2M included, then $2 per 1M
- Realtime: 500 concurrent connections on Pro, 1,000 on Team

Vercel Cost Details

- Pro plan: $20/user/mo with $20 monthly credit
- Bandwidth: 1TB included, then $0.15/GB
- Serverless function execution: 1,000 GB-hours included

- Edge requests: 10M included

Option B: Industry-Standard Stack (AWS/GCP) — Realistic Monthly Costs

This option replaces Supabase with self-managed AWS services and Vercel with AWS Amplify or CloudFront + S3.

Component	AWS Service	Pilot (500 users)	Growth (10K users)	Scale (30K users)	Enterprise (50K+)
Database	RDS PostgreSQL (db.t3.medium → db.r6g.xlarge)	$70	$350	$700	$1,400
Auth	Cognito	$0 (50K MAU free)	$0	$275 (beyond free tier)	$550
API Layer	API Gateway + Lambda	$5	$50	$150	$400
Realtime	AppSync / IoT Core	$20	$100	$250	$500
File Storage	S3 + CloudFront	$5	$30	$80	$200
Frontend Hosting	Amplify / CloudFront + S3	$5	$30	$60	$120
Cron/Scheduling	EventBridge + Lambda	$2	$10	$25	$50
Vector DB (RAG)	RDS pgvector or OpenSearch	$0 (same RDS)	$0 (same RDS)	$150 (OpenSearch)	$300
Redis Cache	ElastiCache	$0 (not needed yet)	$50	$100	$200
Monitoring	CloudWatch + X-Ray	$10	$40	$80	$150
Email	SES	$1	$5	$15	$30
AI (LLM)	Same OpenRouter/OpenAI	$80	$400-700	$1,000-1,800	$2,500-4,000
AI (Embeddings)	Same OpenAI	$5	$40	$100	$200
DevOps Overhead	Engineer time (est.)	$0 (founder)	$2,000-4,000	$4,000-6,000	$6,000-10,000
Dev Tooling	Same as Option A	$331	$331	$331	$331
Total/mo		**~$532**	**~$3,432-5,632**	**~$7,032-10,032**	**~$12,432-18,432**

Note: AWS costs exclude the DevOps engineering time required to manage infrastructure, which is the hidden cost. At growth stage, you need at least a part-time DevOps engineer ($2,000-4,000/mo) that Supabase's managed services eliminate.

Option A vs Option B: Head-to-Head Comparison

Dimension	Option A: Supabase + Vercel	Option B: AWS Self-Managed	Winner
Pilot Cost (500 users)	~$525/mo	~$530/mo	Tie
Growth Cost (10K users)	~$1,850/mo	~$4,530/mo (incl. DevOps)	Option A
Scale Cost (30K users)	~$3,260/mo	~$8,530/mo (incl. DevOps)	Option A
Enterprise Cost (50K+)	~$7,660/mo	~$15,430/mo (incl. DevOps)	Option A
Time to Market	Weeks (managed services)	Months (infrastructure setup)	Option A
DevOps Burden	Near-zero (managed)	High (1-2 engineers needed at scale)	Option A
RLS Security Model	Native PostgreSQL RLS	Must implement in application layer	Option A
Realtime	Built-in (Phoenix Channels)	Must build (AppSync/IoT Core)	Option A
Auth	Built-in (GoTrue)	Cognito (more complex, less flexible)	Option A
Vendor Lock-in	Moderate (Supabase is open-source, can self-host)	Low (AWS services are replaceable)	Option B
Max Scalability Ceiling	~100K users before architectural changes	Virtually unlimited	Option B
Multi-Region	Limited (Supabase regions: US, EU, SG, AP)	Full global coverage	Option B
Compliance Certifications	SOC2 (Team plan), HIPAA (Enterprise)	SOC2, HIPAA, FedRAMP, ISO 27001	Option B
Database Control	Managed (limited tuning)	Full control (instance types, IOPS, replicas)	Option B
Edge Functions	Deno runtime (limited ecosystem)	Lambda (Node.js, Python, full ecosystem)	Option B
Hiring Pool	Smaller (Supabase-specific knowledge)	Massive (AWS is industry standard)	Option B
Disaster Recovery	Managed backups + PITR	Full control (cross-region, custom RPO)	Option B

Recommendation for Edeviser

Stay with Option A (Supabase + Vercel) through the first 30-50 institutions. Here's why:

1. **Cost efficiency at your stage is critical.** At 10K users, Option A costs ~$1,850/mo vs ~$4,530/mo for AWS. That's $32K/year saved — meaningful for a pre-revenue startup.

2. **Speed to market matters more than theoretical scale.** You're targeting Qatar pilot in June 2026. Rebuilding on AWS would add 3-4 months of infrastructure work with zero product progress.

3. **Supabase's RLS is your competitive advantage.** Your entire security model is built on PostgreSQL RLS policies. Migrating to AWS means reimplementing this in application code — a massive regression risk.

4. **The scalability ceiling is far away.** Supabase Pro/Team handles 50K+ users comfortably. You won't hit architectural limits until 100K+ concurrent users across 50+ institutions — likely 2-3 years out.

5. **Exit strategy exists.** Supabase is open-source. If you outgrow managed Supabase, you can self-host the same PostgreSQL + GoTrue stack on AWS/GCP without rewriting application code. This is the best of both worlds.

When to migrate to AWS: When you have 50+ institutions, $500K+ ARR, and can afford a dedicated DevOps team (2-3 engineers). At that point, the cost savings from AWS reserved instances and the compliance requirements of enterprise clients will justify the migration investment.

Constructive Criticism of Current Architecture

What's working well:

- RLS-first security model is genuinely strong and rare in EdTech
- BaaS approach eliminated months of backend boilerplate
- Edge Functions handle complex logic without a separate API server
- The evidence immutability pattern is accreditation-grade

What needs honest attention:

1. **Single Supabase project is a ticking clock.** RLS-based multi-tenancy works, but a single PostgreSQL instance means one bad query from one institution can degrade performance for all. Plan the sharding strategy before you hit 20 institutions, not after.

2. **Edge Functions on Deno are a hiring risk.** Most backend engineers know Node.js/Python, not Deno. As you grow the team, this becomes a friction point. Consider whether critical Edge Functions should be migrated to standard Node.js Lambda functions.

3. **Supabase Realtime has a connection ceiling.** 500 concurrent connections on Pro, 1,000 on Team. With 10,000 students online simultaneously (exam period), you'll hit this. Plan a fallback polling strategy and test it under load.

4. **AI costs will surprise you.** The AI Tutor alone at 10K active students could cost $1,000-2,000/mo. With 50K students, that's $5,000-10,000/mo in LLM API costs. Aggressively cache responses, use cheaper models (DeepSeek) for routine queries, and implement strict rate limiting from day one.

5. **No backend API versioning.** PostgREST auto-generates APIs from your schema. When you change a column name, every client breaks simultaneously. As you onboard multiple institutions on different update cycles, this becomes a deployment risk. Consider a thin API layer for critical endpoints.

6. **The feature surface area is enormous.** 17+ feature modules, 25+ Edge Functions, 30+ database tables. For a startup, this is a lot of surface area to maintain and test. Be ruthless about which features are truly needed for the Qatar pilot vs. which are nice-to-have.

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12. RISK REGISTER

Risk	Probability	Impact	Mitigation
Teacher adoption resistance	High	High	Reduce friction; templates; training videos
Evidence rollup at scale	Medium	High	Index optimization; pre-aggregated views; load testing
Report format varies by body	High	Medium	Configurable templates; start with 2-3 formats
GDPR/FERPA data residency	Medium	High	Supabase region selection; DPA signing; legal review
Gamification gaming	Medium	Medium	XP caps; admin visibility; anomaly detection
Realtime stability at scale	Low	Medium	Fallback polling; graceful degradation
AI hallucinations	Medium	Medium	Human-in-the-loop; confidence thresholds; teacher confirmation
Insufficient AI training data	High	Medium	Activity Logger from day 1; rule-based fallbacks
AI cost overrun	Medium	High	Rate limiting; response caching; cheaper model fallbacks (DeepSeek)
Supabase single-project bottleneck	Medium	High	Monitor query performance; plan sharding at 20 institutions

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13. COMPETITIVE LANDSCAPE

Direct Competitors

- Traditional LMS (Canvas, Blackboard, Moodle) — compliance bolt-ons, no gamification
- OBE-specific tools (Xitracs, AEFIS, Watermark) — compliance only, no engagement
- Gamification platforms (Classcraft, Kahoot) — engagement only, no OBE compliance

Edeviser's Moat

1. **Dual-engine fusion** — Only platform where compliance generates engagement and vice versa
2. **Science-backed design** — 10 research pillars, not arbitrary gamification
3. **AI-native** — RAG-powered tutoring, adaptive quizzes, at-risk prediction built in
4. **Regional focus** — Arabic/Urdu RTL support, HEC/ABET compliance templates
5. **Burnout prevention** — Multi-layered streak recovery, graduated habits, league tiers

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14. OPEN QUESTIONS FOR INVESTOR DISCUSSIONS

#	Question	Status
1	Pricing model: per-student, per-institution, or seat-based?	Open
2	Which accreditation body format for v1.0? (ABET vs HEC vs AACSB vs CNA-Q)	Open
3	Data retention policy for evidence records (GDPR)	Open
4	Multi-tenant architecture transition point (50+ institutions)	Decided: RLS now, revisit later
5	SSO/OAuth integration timeline (Google Workspace, Azure AD)	Phase 2
6	Mobile native app vs PWA strategy	PWA first

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15. ANSWERS TO COMMON INVESTOR QUESTIONS

How does it make money?

SaaS subscription model (B2B). Pricing tiers TBD — likely per-institution with student count tiers. Revenue from: platform license, implementation/onboarding services, premium AI features.

What's the unit economics outlook?

- Low marginal cost per student (Supabase scales well)
- AI costs: OpenRouter API usage (per-token, manageable with caching and rate limits)
- Infrastructure: Vercel + Supabase — predictable, usage-based pricing
- High retention expected due to accreditation lock-in (institutions can't easily switch mid-cycle)

What's the data moat?

- Activity Logger collects behavioral data from day 1
- AI models improve with each institution's data
- Evidence chain creates switching costs (accreditation audit trail)
- Student profiles and learning analytics accumulate over semesters

How do you handle data privacy?

- RLS on every table — security at database layer
- FERPA & GDPR compliance by design
- No PII sent to AI providers
- GDPR data export built in
- Audit trail for all admin actions
- Regional Supabase deployment options

What's the realistic burn rate?

At pilot stage (500 users): ~$525/mo infrastructure + dev tooling (excluding team salaries). At 10K users: ~$1,850/mo. Dev tooling (Claude Code, GitHub Enterprise, CodeRabbit, Figma, Postman, Kiro) adds a fixed $344/mo but replaces the need for additional engineering hires — a single developer with these tools can maintain velocity equivalent to a 3-person team without them.

Edeviser Product Team — April 2026